

**Course Code: BCE-P624**  
**Course Name: MACHINE LEARNING –II**

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| <b>MM: 100</b><br><b>Time: 3 Hr.</b><br><b>L T P</b><br><b>3 0 0</b> | <b>Sessional:30</b><br><b>ESE:70</b><br><b>Credit :3</b> |
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| <b>Prerequisites:</b> | Knowledge of basic machine learning concept, Understanding of Basic Programming Concept and Mathematics (probability and statistics).                                                                                                                                                                                                                                                                                                                                                      |
| <b>Objectives:</b>    | <ul style="list-style-type: none"> <li>To learn the fundamentals of Advanced Machine Learning.</li> <li>To understand basic components of an intelligence system for regression &amp; classification problems.</li> <li>To explore applications and implementation of advanced machine learning.</li> <li>To understand different types of machine learning algorithms, framework and tools.</li> <li>To learn how to use machine learning models to solve real world problems.</li> </ul> |

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| <b>NOTE:</b> | The question paper shall consist of two sections A and B. Section A contains 10 short type questions of 6 marks each and student shall be required to attempt any five questions. Section B contains 8 long type questions of ten marks each and student shall be required to attempt any four questions. Questions shall be uniformly distributed from the entire syllabus. |
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| <b>Module</b>   | <b>Course Content</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>No. of Hours</b> | <b>POs mapped</b>           | <b>PSOs mapped</b> |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-----------------------------|--------------------|
| <i>Module-1</i> | Brief Recap of Fundamentals of Introduction to regression and classification problems, Linear regression in One Variable: Introduction, Gradient Descent Algorithm, Gradient Descent Implementation, Gradient Descent Update Rule for Regression, Data Preparation, Scoring, Surface Plots and Contours, Visualizing Loss function and Gradient Descent Trajectory. Linear regression in multiple variables: Gradient Descent for Multiple Variables, Features and Polynomial Regression. | <b>08</b>           | PO1/<br>PO2                 | PSO1<br>/PSO2      |
| <i>Module-2</i> | Introduction to Logistic Regression: Why use logistic regression? (Binary/Multi-class Classification), Sigmoid Function & Decision Boundary, Applications (Spam Detection, Medical Diagnosis) Cost Function (Log Loss), Gradient Descent Optimization<br>Evaluation Metrics: Confusion Matrix, Precision, Recall, F1-Score, ROC Curve & AUC, Overfitting in Logistic Regression, Multiclass Classification (One-vs-Rest, Softmax)                                                         | <b>08</b>           | PO1/<br>PO2/<br>PO3         | PSO1/<br>PSO2      |
| <i>Module-3</i> | Principal Component Analysis Problem Formulation, PCA Algorithms, K Nearest Neighbors: Distance Metrics, 1-NN algorithm, K-NN algorithm, Weighted KNN algorithm, Issues with high dimensions, high data scarcity and computational complexity, Classification using KNN.                                                                                                                                                                                                                  | <b>08</b>           | PO1/<br>PO2/<br>PO3/<br>PO4 | PSO1/<br>PSO2      |
| <i>Module-4</i> | K-Means Clustering: Intro to Clustering, Intro to Hierarchical Clustering, DBSCAN. Decision Trees and Random Forests: Intro to Decision Trees, Decision Trees implementation, Decision Forests Visualization using, Random Forests Ensembles.                                                                                                                                                                                                                                             | <b>08</b>           | PO1/<br>PO2/<br>PO3/<br>PO4 | PSO1/<br>PSO2      |

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| Module-5           | Introduction and Implementation of Support Vector Machine (SVM).<br>Project Task - End-to-End Machine learning Model Development – Housing Price Prediction, Stock Price Prediction, MNIST Digit Classification, any other. | 08 | PO1/<br>PO2/<br>PO3/<br>PO4/<br>PO5 | PSO1/<br>PSO2 |
| Total No. of Hours |                                                                                                                                                                                                                             | 40 |                                     |               |

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| <b>Learning Outcomes:</b> | <ul style="list-style-type: none"> <li>List various approaches of Advanced Machine Learning and its various tools, framework, data Repositories and datasets.</li> <li>Explain various types of Machine Learning algorithms and machine learning tasks.</li> <li>Identify feature extraction &amp; selection methods, different data types and data pre-processing methods involved in machine learning.</li> <li>Analyze regression &amp; classification problems and various regression classification algorithms</li> <li>Compare various performance metrics for evaluation of Machine Learning Algorithms.</li> </ul> |
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**Suggested books:**

| S. No. | Name of Authors /Books /Publisher                                                                                                                                 |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.     | Introduction to Machine Learning with Python: A Guide for Data Scientists, Andreas C. Müller and Sarah Guido, O'Reilly, 2016.                                     |
| 2.     | Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems, Aurelien Geron, O'Reilly, 2017. |
| 3.     | Elements of Machine Learning, Pat Langley Morgan Kaufmann Publishers, Inc. 1995.                                                                                  |
| 4.     | Machine Learning: The Art and Science of Algorithms that Make Sense of Data, Peter Flach, Cambridge University Press, 2012.                                       |
| 5.     | Pattern recognition and machine learning by Christopher Bishop, Springer Verlag, 2006.                                                                            |
| 6.     | Machine Learning, Tom Mitchell, McGraw Hill, 1997.                                                                                                                |

| CO-PO/PSO MAPPING     |                        |      |      |      |      |      |      |      |      |       |       |       |                                  |       |  |
|-----------------------|------------------------|------|------|------|------|------|------|------|------|-------|-------|-------|----------------------------------|-------|--|
| Course Outcomes (COs) | Program Outcomes (POs) |      |      |      |      |      |      |      |      |       |       |       | Program Specific Outcomes (PSOs) |       |  |
|                       | PO 1                   | PO 2 | PO 3 | PO 4 | PO 5 | PO 6 | PO 7 | PO 8 | PO 9 | PO1 0 | PO1 1 | PO1 2 | PSO 1                            | PSO 2 |  |
| CO1                   | ✓                      | ✓    | ✓    | ✓    |      |      |      |      |      |       |       |       | ✓                                | ✓     |  |
| CO2                   | ✓                      | ✓    | ✓    | ✓    |      |      |      |      |      |       |       |       | ✓                                | ✓     |  |
| CO3                   | ✓                      | ✓    | ✓    |      | ✓    |      |      |      |      |       |       |       | ✓                                | ✓     |  |
| CO4                   | ✓                      | ✓    | ✓    | ✓    |      |      |      |      |      |       |       |       | ✓                                | ✓     |  |
| CO5                   | ✓                      | ✓    | ✓    | ✓    | ✓    |      |      |      |      |       |       |       | ✓                                | ✓     |  |